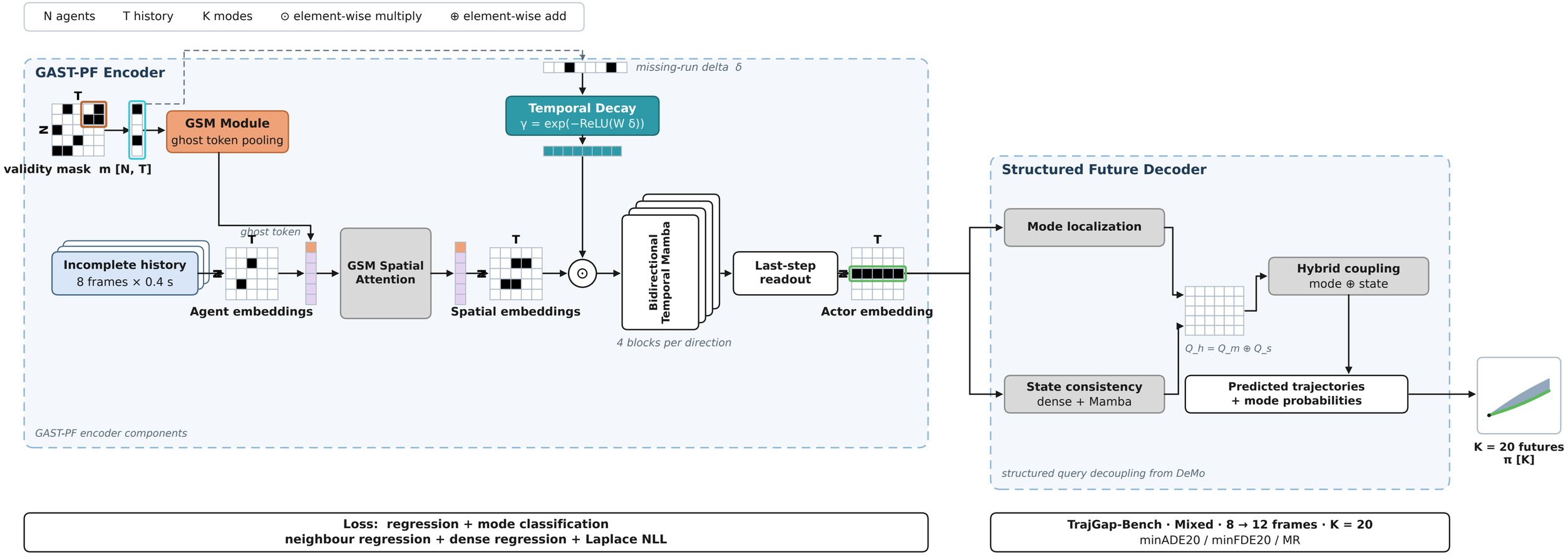
N agents
T history
K modes
⊙ element-wise multiply
⊕ element-wise add
GAST-PF Encoder
T
N
validity mask m [N, T]
GSM Module
ghost token pooling
missing-run delta δ
Temporal Decay
γ = exp(−ReLU(W δ))
ghost token
Incomplete history
8 frames × 0.4 s
T
N
Agent embeddings
GSM Spatial Attention
Spatial embeddings
T
N
⊙
Bidirectional Temporal Mamba
4 blocks per direction
Last-step readout
T
N
Actor embedding
GAST-PF encoder components
Structured Future Decoder
Mode localization
Hybrid coupling
mode ⊕ state
Q_h = Q_m ⊕ Q_s
State consistency
dense + Mamba
Predicted trajectories
+ mode probabilities
K = 20 futures
π [K]
structured query decoupling from DeMo
Loss: regression + mode classification
neighbour regression + dense regression + Laplace NLL
TrajGap-Bench · Mixed · 8 → 12 frames · K = 20
minADE20 / minFDE20 / MR